AI for Manufacturing GRCC Training

AI for Workforce

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Welcome!

Get all the content at:
https://github.com/grcc-resendiz/ai-manufacturing-muskegon

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Introductions



Jonnathan is an Assistant Professor at the Computer Information Systems Department and the Faculty Director of the AI Incubator at GRCC. He has a M.S. in Computer Science from the University of Texas at Dallas.

Clayton Corrello teaches in the Business/Computer Information Systems Department at GRCC, including AI courses where students run local LLMs with LM Studio. He earned his B.S. from Northern Michigan University and his M.S. from Grand Valley State University.

Day 1 Agenda

- **0:00** Welcome and introductions
- **0:10** The state of manufacturing: what keeps you up at night?
- **0:30** Prompting: hallucinations, safe use, and the RTFC framework
- **0:55** AI projects: build one for your job
- **1:55** Break
- **2:00** Local LLMs with LM Studio (Clayton)
- **2:55** Wrap up and questions

Introduction to AI

What is Artificial Intelligence?

- Artificial intelligence (AI) is a machine's ability to understand, reason, and learn, aiming to achieve human-like intelligence.

- AI refers to developing computer systems that can perform tasks that typically require human intelligence, such as visual perception, speech recognition, decision-making, and language translation.

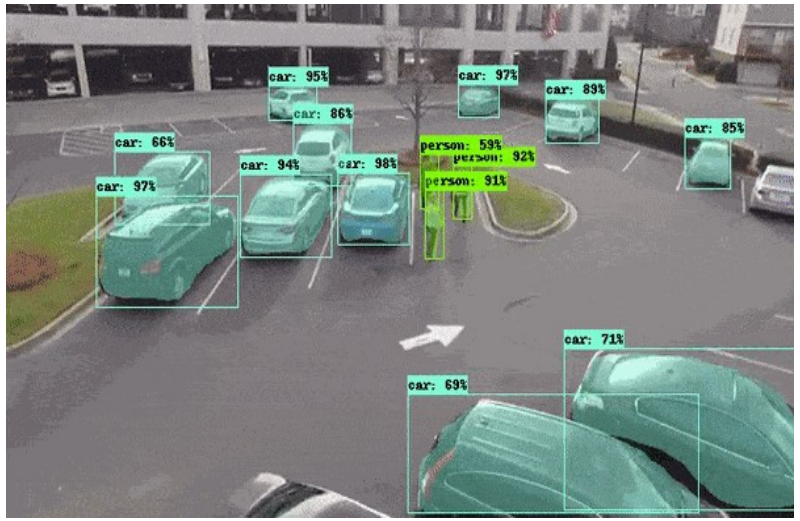


Reference =Goddard, W. (2021, September 9). Where is Artificial Intelligence Used Today? ITChronicles. https://itchronicles.com/artificial-intelligence/where-is-ai-used-tod

Introduction to AI

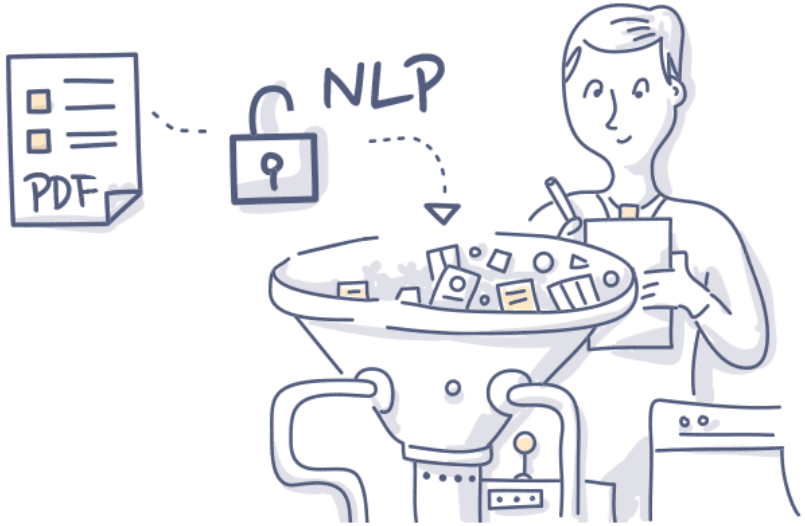
Domains of AI

Three main Domains in Artificial Intelligence are:



Computer vision

The ability of machines to see the world



Natural Language Processing

The ability of machines to understand human language



ML/Statistical Data

The ability of machines to understand numbers

What You Will Do in 3 Days

- **Day 1:** Prompt AI tools well and build a reusable AI project for your job
- **Day 2:** Train a model to spot defective cast parts, then vibe code a simple app
- **Day 3:** Predict machine failures from sensor data with Orange
- **Day 3:** Plan your first AI pilot

Manufacturing Discussion

What Keeps You Up at Night?

Problems faced in the manufacturing Industry

- Downtime
- Quality
- Scrap
- Hiring
- Training
- Scheduling
- Safety



The Manufacturing Reality (2024 to 2033 Outlook)

- Up to 3.8 million manufacturing jobs to fill (2024 to 2033)
- About 1.9 million could go unfilled
- Workforce shortage is becoming the primary constraint on growth
  - This is not a future problem.
  - It is already happening.

Two Big Challenges

- Workforce shortage
- Technology adoption gap
  - We do not have a technology problem.
  - We have an adoption problem.

AI adoption in Manufacturing as a Solution

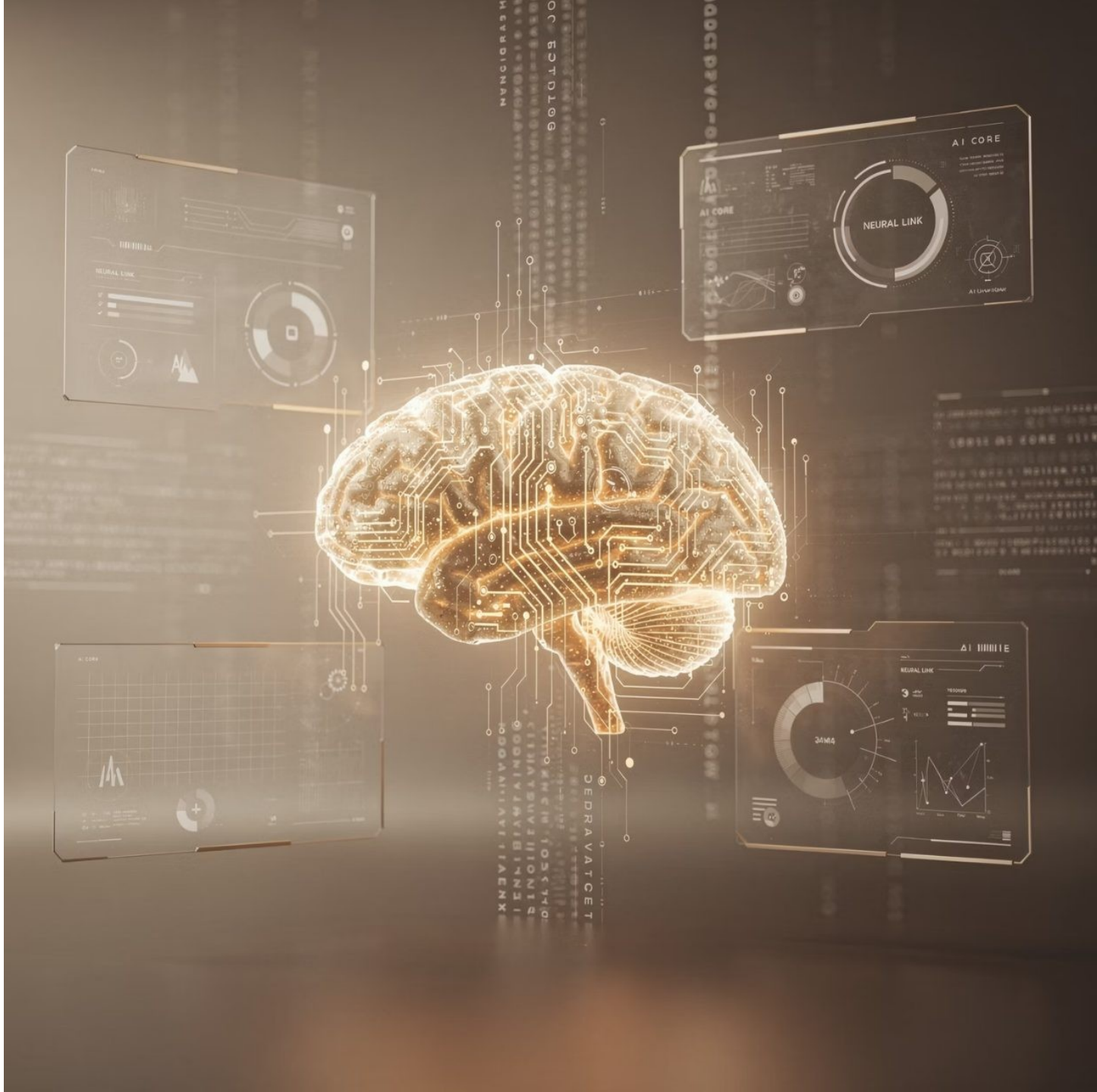
- AI amplifies learning
  - AI can be used for learning other technologies
- There's no better time to learn how to code and implement tech
- If we invest in AI training we could solve both problems
- There's no one coming to save us, we need to implement solutions ourselves



Image Source- M. (2022a, May 25). Pros And Cons of AI In Manufacturing. Data Science Central. https://www.datasciencecentral.com/pros-and-cons-of-ai-in-manufacturing/

Prompt Engineering

Get all the content at: https://github.com/grcc-resendiz/ai-manufacturing-muskegon



Overview

Prompt Engineering

- Learn how to effectively prompt large language models (LLMs) like ChatGPT, Gemini, and Claude.
- Familiarize yourself with the RTFC framework.

AI Projects

- Build a project: saved instructions plus your documents, in the AI tool you already use.
- Goal: Package your instructions and documents once so the AI gives consistent, tailored answers every time.

Beware of Hallucinations!

> **A hallucination is when a model invents details, sources or data that do not exist.**

Examples:

Incorrect Settings

Giving incorrect machine settings, tolerances, or specifications

Nonexistent Policies

Referencing non-existent company policies or outdated SOPs

Fake Citations

Fake citations or sources (e.g., "OSHA requires..." when it doesn't)

Incorrect Claims

Incorrect claims about recent events, regulations, or recalls

Wrong Calculations

Confidently providing incorrect measurements, conversions, or calculations

Use AI Safely at Work

- Follow your company AI policy first.
- Do not paste confidential drawings, customer data, or personal information into public AI tools.
- Check every number, setting, and spec against the source before you act on it.
- Need to keep data in house? Local models run on your own computer. Clayton shows you how today.

Prompt Fundamentals

One strategy for effectively prompting LLMs is the RTFC Framework:

1

Role

Describe *who* the model should act as. Ex: You are a maintenance technician

2

Task

Describe *what* you want done: Ex: Identify possible causes of this machine issue.

3

Format

Specify *how* you want the answer presented. Ex: List the top 3 likely causes.

4

Constraints

List *limits* or *rules*: Ex: Only use the provided document.

RTFC Example

Prompt:

> You are a production supervisor preparing a shift handoff.
> Summarize the following update.
>
> Provide your answer in 3 sections:
> - Issue
> - Impact
> - Required follow-up
>
> Use bullet points and keep each section concise.
>
> Message:
> “During the morning shift, the assembly line experienced intermittent stops due to a sensor issue. Operators had to manually reset the system several times. This caused about 45 minutes of total downtime. The issue seems to happen more frequently when the line is running at higher speeds. No permanent fix has been implemented yet, and maintenance has been notified.”

Projects: Reusable AI Instructions

Session Context

Typically, you must provide context within each interaction.

Reusable Setup

A project stores your instructions and files once. Every chat in the project uses them.

Built In Examples

Add sample inputs and outputs so every answer follows the same pattern.

Result

Saves time and improves consistency for repetitive workplace tasks

Examples



Standard Operating Procedures (SOPs)

Load SOPs, manuals, and internal documents so the project can answer questions and guide employees consistently



Troubleshooting & Support

Provide past issues, solutions, and logs so the project can help identify likely causes and next steps



Training & Onboarding

Include training materials and examples so the project can explain processes to new employees in a consistent way



Reporting & Communication

Load examples of shift reports or updates so the project can help generate clear, standardized summaries

Where to Build Your Project

- **Claude:** Projects, on every plan (free allows 5 projects)
- **ChatGPT:** Projects, on every plan (free allows 5 files per project)
- **Microsoft Copilot:** Notebooks, with a Copilot or Copilot Chat license
- **Gemini:** Gems on work accounts; skills on personal accounts
- **Coming next:** skills, the same instructions in a file that goes with you

Anatomy of a Project

Instructions (RTFC)

- Role, task, format, and constraints, written once.
- Example: a mill maintenance helper for the Bridgeport knee mill.
- The same text works in every tool. Keep a copy.

Files (optional)

- Manuals, SOPs, checklists, past shift reports.
- Save documents as PDF; every tool reads PDF.
- Free ChatGPT allows 5 files per project.

Build It: Claude and ChatGPT

Claude (free plan works)

- Projects > + New Project, and name it.
- Set project instructions: paste yours, then Save.
- Click + next to project knowledge to add files.

ChatGPT (free plan works)

- New project in the sidebar, and name it.
- … (top right) > Project settings: paste your instructions.
- Add your files to the project, then chat inside it.

Build It: Copilot and Gemini

Microsoft Copilot (Notebooks)

- m365.cloud.microsoft > Notebooks > New notebook.
- Add your files under References, then Create.
- … > Instructions: paste your instructions, then Save.

Gemini (Gems or skills)

- Work account: Gems > New Gem. Paste instructions; Knowledge > Add files.
- Personal account: Settings > Skills > Create manually.
- Test with 3 real questions, then fix the instructions.

Bonus: Turn It Into a Skill

- Add a name and a one line description to the top of your instructions.
- Save it as SKILL.md and upload it in Claude (Customize > Skills) or Gemini (Settings > Skills).
- Gems and custom GPTs are turning into skills, and this file goes with you to any tool.
- Download the demo: skill/mill-maintenance-helper.zip on the course GitHub page.

Local LLMs with LM Studio

Led by Clayton

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Next Session: Computer Vision

- You will train a defect detection model.
- You will test it live.
- You will see how AI sees defects.

Questions?


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